

Prelim Prep

Multiple choice questions and identification of commands
85 points

Terms and Def

Introduction to Systems Administration and Maintenance

System Administration and Maintenance are described as crucial aspects of managing and ensuring the smooth operation of computer systems, networks, and servers within an organization.

This means that system administration is not optional—it is necessary for keeping IT infrastructure operational and reliable.

The slides emphasize that these tasks involve various activities aimed at:

- Optimizing **performance**
- Enhancing **security**
- Resolving issues
- Keeping the IT infrastructure running efficiently

From this, we understand that system administration is both preventive and corrective. It prevents problems by maintaining systems properly and corrects issues when they arise.

Importance of System Administrators in Organizations

The presentation clearly states that **System Administrators (SysAdmins)** play a critical role in the smooth operation, security, and reliability of an organization's IT infrastructure.

This means organizations depend heavily on SysAdmins because:

- Systems must stay operational.
- Data must remain secure.
- Infrastructure must be reliable at all times.

Without system administrators, organizations would struggle with downtime, security breaches, and system failures.

Role and Responsibilities of a System Administrator

A **system administrator**, often referred to as a **sysadmin**, plays a pivotal role in managing and maintaining technological infrastructure.

Primary Objective

The primary objective of a sysadmin is to ensure:

- **Reliability**
- **Security**
- **Efficient operation**

of:

- Computer systems
- Networks
- Servers

This objective defines the entire purpose of the role.

Core Role Description

The sysadmin is responsible for:

- Overseeing the day-to-day functioning of IT systems
- Troubleshooting issues
- Implementing security measures
- Supporting end-users

Detailed Responsibilities

The slides list specific responsibilities:

1. **System Setup and Configuration**
Installing and configuring systems properly so they function as intended.
2. **User Account Management**
Creating, managing, and maintaining user accounts.
3. **Network Management**
Ensuring the network infrastructure works correctly.
4. **System Monitoring and Performance Optimization**
Watching system behavior and improving performance when needed.
5. **Security Implementation**
Applying security measures to protect systems and data.
6. **Data Backup and Recovery**
Ensuring backups are available and data can be restored when necessary.
7. **Troubleshooting and Issue Resolution**
Identifying and fixing problems in systems.
8. **Hardware and Software Maintenance**
Maintaining both physical and software components.
9. **Automation and Scripting**
Using scripts and automation to improve efficiency.
10. **Documentation and Reporting**
Recording configurations, processes, and system changes.

Importance of System Administrators

The slides further detail why sysadmins are important:

1. **Ensuring System Availability and Uptime**
Systems must remain accessible and operational.
2. **Maintaining Security and Protecting Data**
Data protection is a core responsibility.
3. **Managing Users and Resources**
Users must have proper access and system resources must be allocated efficiently.
4. **Supporting Business Operations**
IT infrastructure directly supports organizational activities.
5. **Preventing Data Loss and Ensuring Recovery**
Backup systems prevent permanent data loss.
6. **Cost Efficiency and Resource Optimization**
Efficient systems reduce unnecessary expenses.
7. **Compliance and Policy Enforcement**
Systems must follow organizational and legal policies.

Compatibility and Integration in Creating a Seamless Computing Environment

To create a seamless computing environment, multiple aspects must be considered:

- **Hardware and Software Harmony** – Ensuring compatibility between hardware and software.
- **System Stability** – Systems must operate without frequent crashes.
- **Optimized Performance** – Systems must function efficiently.
- **Interoperability** – Different systems and components must work together.
- **User Experience** – Systems should be usable and reliable.
- **Cost-Efficiency** – Avoid unnecessary spending.
- **Security Integration** – Security must be integrated into systems.
- **Scalability** – Systems should grow when needed.
- **Efficient Maintenance** – Maintenance should be manageable.
- **Adaptability to Change** – Systems must adjust to evolving technology.

These aspects ensure computing environments remain functional and adaptable.

Career Paths

The presentation identifies several related roles:

- **Network Administrator**
- **Web Administrator**
- **Database Administrator**
- **Security Administrator**
- **System Administrator**

These represent specialized directions within IT infrastructure management.

Skills and Qualities of a System Administrator

Technical Skills

- Proficiency in **operating systems**
- Knowledge of **networking**
- Strong **troubleshooting** ability

Communication Skills

- Ability to interact with technical and non-technical users

Problem-Solving Abilities

- Identifying and resolving issues promptly

Adaptability

- Keeping pace with evolving technologies

These skills reflect both technical competence and professional behavior.

Types of Systems

The slides identify three main types:

1. Desktop Systems

- Used by **individual users**
- Examples: office PCs, laptops
- Focus on **usability and application support**

2. Server Systems

- Provide services to **multiple users**
- Examples: file servers, web servers, database servers
- Focus on **uptime, performance, and security**

3. Cloud Systems

- Hosted over the internet
- Examples: AWS, Azure, Google Cloud
- Features:
 - **Scalability**
 - **Remote access**
 - **Pay-as-you-go**

A comparison table further summarizes:

- Desktop → Single user, local
- Server → Multiple users, data center
- Cloud → Multiple users, internet-based

Operating Systems and Best Practices

Introduction to Operating Systems

An **Operating System (OS)** is software that:

- Manages hardware resources
- Manages software resources
- Provides services to applications

Main Functions of an OS

- **Process management**
- **Memory management**
- **File system management**
- **Device management**
- **User interface**

These functions define how the OS controls and coordinates system resources.

Client vs Server Operating Systems

Client Operating Systems

- Designed for end users
- Provide user-friendly interfaces
- Examples:
 - Windows 10 / 11
 - Ubuntu Desktop
 - macOS

Server Operating Systems

- Designed to provide services to clients
- Optimized for performance and security
- Examples:
 - Windows Server
 - Ubuntu Server
 - Red Hat Enterprise Linux

Feature Comparison

Feature	Client OS	Server OS
Users	Single/few	Multiple
Interface	GUI	GUI / CLI
Stability	Moderate	High
Cost	Lower	Higher

System Administration Ethics

Ethical responsibilities include:

- **Responsible use of system access**
- **Respect for user privacy**
- **Confidentiality of data**
- **No unauthorized access or misuse**

Ethics ensure trust and integrity in system administration.

Best Practices in System Administration

The slides list the following best practices:

- **Strong password policies**
- **Regular system updates and patching**
- **Data backups**
- **Proper documentation**
- **Least privilege principle**

These practices improve security and system reliability.

Virtual Machines

Learning Objectives

By the end of the lesson, students should be able to:

- Explain **virtualization**
- Define a **Virtual Machine (VM)**
- Identify components of a VM
- Explain why VMs are important in system administration

What Is Virtualization?

Virtualization is defined as the process of creating a virtual version of a computer system.

It allows one physical computer to run multiple operating systems at the same time.

The slides contrast:

Traditional:

- One OS = One machine

Virtualized:

- Multiple OS = One machine

This highlights efficiency and resource utilization.

Why Virtualization Matters

Virtualization is important because:

- It is used in **data centers**
- It is used in **cloud platforms**
- It reduces **hardware cost**
- It enables **testing, development, and isolation**

Isolation is particularly important because it allows safe experimentation without affecting other systems.

What Is a Virtual Machine?

A **Virtual Machine (VM)** is a software-based computer that runs inside a physical computer.

Each VM has its own:

- **CPU**
- **Memory (RAM)**
- **Storage**
- **Network interface**
- **Operating system**

This means each VM functions like an independent computer.

Key VM Components

1. **Host OS**
The operating system installed on the physical machine.
2. **Guest OS**
The operating system installed inside the virtual machine.
3. **Hypervisor**
The software that manages virtual machines.

The hypervisor controls resource allocation and VM management.

Types of Hypervisors

Type 1 (Bare-metal Hypervisor)

- Runs directly on hardware
- Provides best performance
- High security
- Resource efficiency
- Used in enterprise and cloud environments

Examples:

- VMware ESXi
- Microsoft Hyper-V Server

Type 2 (Hosted Hypervisor)

- Runs on top of an existing OS
- Common for learning, development, and testing

Examples:

- VirtualBox
- VMware Workstation

Benefits of Virtual Machines for SysAdmins

The slides list several key benefits:

- **Isolation**
If one VM crashes, others are not affected.
- **Testing**
Safely test updates and configurations.
- **Snapshots**
Roll back to previous states.
- **Scalability**
Quickly deploy new servers.
- **Cloud foundation**
VMs form the base of AWS, Azure, and GCP.

What Is a Server?

A **server** is defined as a computer system that provides services, resources, or data to other computers called **clients** over a network.

This means a server does not primarily serve a single user. Instead, it exists to respond to requests coming from multiple client machines.

For example:

- A file server provides files to users.
- A database server provides database services.
- A DHCP server assigns IP addresses.

The focus of a server is service delivery rather than user interaction.

Key Characteristics of a Server

The slides highlight important characteristics:

1. **Designed to run continuously (24/7)**
Servers are expected to stay operational at all times without frequent shutdowns.
2. **Optimized for reliability and performance**
Reliability ensures the system does not frequently fail.
Performance ensures it handles requests efficiently.
3. **Handles multiple client requests simultaneously**
Servers are built to serve many users at the same time.
4. **Focus on services, not user interface**
Unlike desktop systems, servers prioritize backend service functionality over graphical interface.

Desktop vs Server Comparison

The slides compare desktops and servers to clarify differences.

Feature	Server	Desktop
Purpose	Provide services	Personal use
Uptime	Always on	On demand
Users	Multiple	Single
Interface	CLI-focused	GUI-focused

Explanation of Comparison:

- **Purpose**
A desktop system is designed for personal productivity.
A server is designed to provide services to other systems.
- **Uptime**
Servers remain active continuously.
Desktops are turned on only when needed.

- **Users**
Servers handle many users.
Desktops are generally for one user.
- **Interface**
Servers are often **CLI-focused (Command Line Interface)** because graphical interfaces are unnecessary and may consume extra resources.
Desktops are **GUI-focused (Graphical User Interface)** for ease of use.

Types of Servers

The presentation lists the following types:

1. **File Server**
Stores and manages files accessible over a network.
2. **Database Server**
Manages and provides access to databases.
3. **DHCP Server**
Assigns IP addresses automatically to devices in a network.
4. **Proxy Server**
Acts as an intermediary between clients and other servers.

Each server type exists to fulfill a specific network service function.

Best Practices in Server Deployment

The slides provide structured best practices for deploying servers properly.

1. Plan Before Deployment

Before setting up a server:

- Define the **purpose of the server** (web, database, file, DHCP, etc.)
- Identify **system requirements** (CPU, RAM, storage)
- Choose the appropriate **Operating System and version**

Planning ensures that the server is built correctly from the start.

2. Use Stable and Supported Operating Systems

- Prefer **Long-Term Support (LTS)** versions (e.g., Ubuntu Server LTS)
- Avoid unsupported or end-of-life releases

Using supported systems ensures security updates and stability.

3. Minimize Installed Components

- Install only required packages and services
- Avoid unnecessary GUI installations on servers

Minimal installation:

- Reduces security risks
- Improves performance
- Simplifies maintenance

4. Secure the Server from the Start

Security must be implemented immediately.

- Change default passwords
- Use strong authentication
- Configure firewalls (e.g., UFW)
- Secure SSH access (disable root login if possible)

This prevents unauthorized access early in deployment.

5. Apply Updates and Patches Immediately

- Update the system after installation
- Enable automatic security updates when appropriate

Updates fix vulnerabilities and improve stability.

6. Configure Proper User and Permission Management

- Use individual user accounts
- Apply the **principle of least privilege**
- Use **sudo** for administrative access

The **least privilege principle** ensures users only receive necessary permissions.

7. Implement Backup and Recovery Early

- Plan regular backups
- Test backup restoration
- Store backups securely

Backups protect against data loss and system failure.

8. Monitor System Performance and Logs

- Monitor CPU, memory, disk usage
- Review log files regularly
- Set alerts for critical events

Monitoring ensures early detection of problems.

9. Document Everything

- Record installation steps
- Save configuration details
- Keep change logs

Documentation supports maintenance, troubleshooting, and continuity.

Summary of Server Deployment

A properly deployed server should be:

- **Planned**
- **Secure**
- **Minimal**
- **Documented**
- **Maintainable**

Linux File Systems

Linux File System Structure

The Linux file system has three structural layers:

1. Logical File System

The **Logical File System** acts as the interface between user applications and the file system.

It:

- Handles operations like **open, read, write, and close**
- Performs **security checks**
- Enforces **permissions and file access control**
- Ensures applications interact with files consistently

It hides storage complexity from applications.

2. Virtual File System (VFS)

The **Virtual File System (VFS)**:

- Provides a common interface for different file systems (ext4, XFS, FAT32, NTFS, etc.)
- Allows Linux to use multiple file systems simultaneously
- Acts as an abstraction layer
- Makes file operations uniform regardless of file system type

VFS ensures compatibility and consistency.

3. Physical File System

The **Physical File System**:

- Directly interacts with hardware and disk storage
- Manages **data blocks**
- Manages **inodes**
- Handles physical memory allocation
- Writes and retrieves data efficiently
- Ensures reliable storage and error handling

This layer is responsible for actual data storage.

Introduction to Linux File System

The **Linux File System** is a structured method of storing and organizing data.

It uses:

- A **hierarchical directory format**
- A starting point called the **root directory (/)**

All files and directories branch from the root.

Everything Is a File in Linux

In Linux:

- Files
- Directories
- Devices
- Processes
- Network connections

are treated as files.

Linux does not rely on file extensions like .exe or .txt to determine file type.

File Types in Linux

Linux identifies file types using file attributes, shown by the first character in a list.

1. Regular File (-)

Stores data or program code.

Examples:

- Text files (.txt, .conf)
- Executable programs
- Images and videos

2. Directory (d)

Contains other files and directories.

3. Symbolic Link (l)

A shortcut or reference to another file.
Similar to Windows shortcuts.

4. Character Device File (c)

Transfers data one character at a time.

Examples:

- Keyboard
- Mouse
- Terminal

Located in **/dev**

5. Block Device File (b)

Transfers data in blocks.

Examples:

- Hard drives
- USB drives

Located in **/dev**

6. Socket File (s)

Used for **inter-process communication**.
Common in network services.

7. Named Pipe / FIFO (p)

Temporary communication channel between processes.

Why File Types Matter for SysAdmins

Understanding file types:

- Prevents accidental deletion of system files
- Helps identify device files
- Ensures correct permissions
- Is critical for troubleshooting and security

Single Root Structure

Linux uses a **single directory tree** starting from the root directory */*.

Unlike Windows (C:, D:), Linux mounts all storage devices into this single directory structure.

This means everything connects under one root.

Linux Directory Structure

Important directories include:

- */* – Root directory
- */home* – User home directories
- */etc* – System configuration files
- */var* – Logs and variable data
- */bin* – Essential user commands
- */usr* – User applications
- */tmp* – Temporary files
- */dev* – Device files
- */proc* – System and process information

Each directory serves a specific function in the system.

Linux File Systems (Historical and Modern)

The slides outline major Linux file systems:

1. **ext (Extended File System)** – First Linux file system (1992)
2. **ext2** – Non-journaling, improved performance
3. **Xiafs** – Alternative to ext2, discontinued
4. **ext3** – Added **journaling** for reliability
5. **JFS (Journaled File System)** – Developed by IBM
6. **ReiserFS** – Advanced features, later replaced
7. **XFS** – 64-bit journaling, supports large storage
8. **SquashFS** – Compressed, read-only file system
9. **Reiser4** – Successor to ReiserFS, limited adoption
10. **ext4** – Backward compatible, supports large files, default in many distributions
11. **btrfs (B-tree File System)** – Snapshots, drive pooling, self-healing
12. **bcache** – Copy-on-write, encryption, compression
13. **Others (NTFS, exFAT)** – Cross-platform compatibility, limited Unix-style permissions.

Linux/Windows Commands

Windows CLI Commands

User Management

net user

Purpose: Manage local user accounts (create, modify, delete, reset passwords).

Common Operations:

Add a new user:

```
net user {username} {password} /add
```

Delete a user:

```
net user {username} /delete
```

Change a user's password:

```
net user {username} *
```

2. Group Management

net localgroup

Purpose: Manage local security groups and memberships.

Create a new group:

```
net localgroup {groupname} /add
```

Add a user to a group:

```
net localgroup {groupname} {username} /add
```

Linux CLI Commands

User Management

adduser

Purpose: Create a new user account (with home directory).

```
sudo adduser {username}
```

deluser

Purpose: Remove a user account.

```
sudo deluser {username}
```

passwd

Purpose: Change a user's password.

```
sudo passwd {username}
```

groupadd

Purpose: Create a new group.

```
sudo groupadd {groupname}
```

usermod

Purpose: Modify user account attributes (groups, shell, etc.).

Add user to a group:

```
sudo usermod -aG {groupname} {username}
```

chmod (Change Mode)

Purpose: Modify file or directory permissions.

Who:

u → user (owner)

g → group

o → others

a → all (user, group, others)

Operators:

+ → add (set) permission

- → remove (revoke) permission

= → assign exact permission (reset and set)

Permissions:

r → read **w** → write **x** → execute

Symbolic mode example:

```
chmod u+rx,g=rw,o-x {filename}
```

*This **adds** read and execute permission on the owner, **set(override)** read and write permission on the group, and **revoke** execute permission for others.*

Numeric mode example:

```
chmod {permission_number} {filename}
```

Permission breakdown example (752):

- Owner: 7 → read, write, execute
- Group: 5 → read, execute
- Others: 2 → write

chown (Change Ownership)

Purpose: Change file owner and/or group.

Change owner and group:

```
sudo chown {newowner}:{newgroup} {filename}
```

Change group only:

```
sudo chown :{newgroup} {filename}
```

Storage Management

In Linux, storage devices (hard drives, USB drives, partitions) must be **mounted** before you can access their files. Mounting connects a device to a directory in the filesystem.

mount

Purpose: Attach a storage device to the filesystem.

```
sudo mount {device} {mount_point}
```

- `{device}` → The disk or partition (e.g., `/dev/sdb1`)
- `{mount_point}` → The directory where it will be accessible (e.g., `/mnt/usb`)

Example:

```
sudo mount /dev/sdb1 /mnt/usb
```

This makes the contents of `/dev/sdb1` accessible inside `/mnt/usb`.

umount

Purpose: Safely detach a mounted storage device.

```
sudo umount {mount_point}
```

Example:

```
sudo umount /mnt/usb
```

Always unmount before physically removing a USB drive to prevent data corruption.

Quiz 1

Quiz Results Summary

Score: 14 / 15 (93.33%)

Time: 10 minutes 10 seconds

Question 1

Question:

The company needs to regularly back up its transaction records and ensure recovery in case of failure.

Answer:

Database Administrator

Clarification:

Database backups and recovery planning fall under database management responsibilities.

Question 2

Question 1:

A company's customer records are taking too long to load, and indexes need to be optimized to improve query performance.

Answer:

Database Administrator

Clarification:

Index optimization directly affects database query performance.

Question 2:

An organization wants to deploy a new company website and configure the web server to handle user requests efficiently.

Answer:

Web Administrator

Clarification:

Web server deployment and configuration are core web administration tasks.

Question 3:

Employees in different departments cannot access the internal system due to incorrect IP addressing and subnet configuration.

Answer:

Network Administrator

Clarification:

IP addressing and subnetting are network infrastructure responsibilities.

Question 4:

Sensitive employee data was accessed without authorization, and an investigation into access logs and permissions is required.

Answer:

Security Administrator

Clarification:

Unauthorized access investigations fall under security monitoring and auditing.

Question 3

Question 1:

Staff members experience slow access to online tools at certain times of the day. Monitoring shows congestion patterns that vary depending on location and usage.

Answer:

Network Administrator

Clarification:

Network congestion analysis is part of traffic and bandwidth management.

Question 2:

A system audit reveals that several machines are communicating with unknown external addresses, and access rules may need adjustment to reduce risk.

Answer:

Security Administrator

Clarification:

Managing firewall rules and mitigating external threats is a security function.

Question 3:

A department reports that after recent changes, some employees can view records they previously could not, while others lost access entirely.

Answer:

Database Administrator

Clarification:

Access control changes affecting records typically involve database permissions.

Question 4:

A company application handles user login data, and management wants to ensure that transmitted information cannot be intercepted or altered.

Answer:

Security Administrator

Clarification:

Protecting transmitted data requires encryption and secure communication controls.

Question 5:

Two office branches are connected, but users report inconsistent access to shared services depending on where they log in from.

Answer:

Network Administrator

Clarification:

Inter-branch connectivity and routing consistency are networking responsibilities.

Question 6:

System reports show repeated access attempts from multiple sources, and current protection measures do not appear sufficient.

Answer:

Security Administrator

Clarification:

Monitoring suspicious access attempts is part of intrusion detection and response.

Question 7:

An organization plans to reorganize how its online content is delivered while keeping services available to users throughout the transition.

Answer:

Web Administrator

Clarification:

Content delivery restructuring while maintaining uptime involves web service management.

Question 8:

Historical records are growing rapidly, and system performance is beginning to degrade due to how information is stored and retrieved.

Answer:

Database Administrator

Clarification:

Storage structure and retrieval optimization are database management tasks.

Question 9:

Remote staff can connect to internal systems, but some sessions fail unexpectedly, raising concerns about reliability and secure access.

Answer:

Network Administrator

Clarification:

Remote connectivity reliability typically involves VPN and network configuration management.

Question 4

Question:

During peak business hours, users report that an online system becomes unavailable even though the server is running. Logs show repeated service restarts and configuration warnings.

Answer:

Web Administrator

Clarification:

Service restarts and web server configuration issues are web administration concerns.

EXAM RESULTS

Correct Answers

1. Which of the following is a primary responsibility of a system administrator?

Answer: Managing server performance and security

2. Which role focuses on protecting systems from unauthorized access and cyber threats?

Answer: Security Administrator

3. Which of the following is an example of a Linux distribution?

Answer: Ubuntu

4. What is the main difference between a Client OS and a Server OS?

Answer: Server OS is designed to manage resources and services for multiple users

5. A virtual machine primarily allows you to:

Answer: Run multiple operating systems on one physical machine

6. Which component is MOST important for virtualization performance?

Answer: CPU

7. During OS installation, partitioning is important because it:

Answer: Divides the disk into manageable sections

8. What is the purpose of setting a hostname during server installation?

Answer: To identify the server on a network

9. A system administrator is responsible for system updates and patch management.

Answer: True

10. Desktop operating systems are optimized for providing network services to hundreds of users simultaneously.

Answer: False

11. Virtual machines require a hypervisor to run.

Answer: True

12. Skipping system updates after OS installation is considered a best practice.

Answer: False

13. "The software layer that allows virtual machines to run is called a ____."

Answer: hypervisor

14. "The command-line interface in Linux is commonly referred to as the ____."

Answer: Shell

15. "The process of preparing a storage device for use by creating file systems is called ____."

Answer: Formatting



Practical Commands

16. Directory Structure

```
sudo mkdir -p /data/innovatech/{engineering,sales,finance,shared,confidential}
```

17. Users and Groups

Create Groups

```
sudo groupadd enggrp  
sudo groupadd salesgrp  
sudo groupadd financegrp
```

Create Users and Assign to Groups

```
sudo useradd -m -G enggrp eng1  
sudo useradd -m -G salesgrp sales1  
sudo useradd -m -G financegrp fin1
```

Set Passwords

```
sudo passwd eng1  
sudo passwd sales1  
sudo passwd fin1
```

18. Ownership and Permissions

Set Group Ownership

```
sudo chown :enggrp /data/innovatech/engineering  
sudo chown :salesgrp /data/innovatech/sales  
sudo chown :financegrp /data/innovatech/finance
```

Set Permissions (Department Folders)

```
sudo chmod 770 /data/innovatech/engineering  
sudo chmod 770 /data/innovatech/sales  
sudo chmod 770 /data/innovatech/finance
```

Shared Folder

```
sudo chmod 777 /data/innovatech/shared
```

Confidential Folder

```
sudo chmod 700 /data/innovatech/confidential
```

19. What does permission 770 mean?

Answer: Owner and group can read/write/execute, others have no access

20. Who can access **drwxr-x---** root enggrp engineering?

Answer: Root and members of enggrp

21. Can sales1 enter this folder?

Answer: No, because others have no permissions

22. Which command correctly gives full access to enggrp only?

Answer: **chmod 770 engineering**

23. What does r-x represent?

Answer: Read and execute

24. Why can't eng1 create files in `drwxr-xr-x`?

Answer: Group has no write permission

25. Why can't eng1 create a file?

Answer: The group lacks write permission

26. Which command fixes the issue correctly?

Answer: `chmod 770 engineering`

27. After fixing, what should the permission look like?

Answer: `drwxrwx---`

Midterm Prep

Study module 6 and 7 only.

MCQ

True or false

And Identification based on the commands used during activities.

Quiz 2

Part 1: Identification

1. A program that runs in the background to provide functionality.
Answer: Service
2. A background process that starts automatically and waits for requests.
Answer: Daemon
3. The command-line tool used to manage services on Ubuntu Server.
Answer: `systemctl`
4. The protocol that automatically assigns IP addresses, subnet masks, gateways, and DNS servers.
Answer: DHCP
5. A device that forwards DHCP messages between clients and servers across different network segments.
Answer: DHCP Relay Agent
6. The specific duration for which an IP address is allocated to a DHCP client.
Answer: DHCP Lease
7. A range of IP addresses from which DHCP servers can assign addresses to clients.
Answer: DHCP Scope
8. A feature that allows administrators to assign specific IP addresses to devices based on their MAC addresses.
Answer: DHCP Reservation
9. The command used to check current network interface configurations.
Answer: `ip a` (or `ifconfig`)
10. The configuration file path commonly edited to set a static IP in Netplan.
Answer: `/etc/netplan/50-cloud-init.yaml`
11. The command to apply new network configurations in Netplan.
Answer: `sudo netplan apply`
12. A conceptual framework that standardizes networking functions into seven abstraction layers.
Answer: OSI Model
13. The protocol suite named after its two most important protocols: TCP and IP.
Answer: TCP/IP

14. The process of selecting the best path for network traffic from source to destination.
Answer: Routing
15. A device that forwards packets between networks based on destination IP addresses.
Answer: Router
16. Each router maintains this list of known networks and next-hop interfaces.
Answer: Routing Table
17. A type of routing where administrators manually configure the routing table.
Answer: Static Routing
18. A type of routing where routers exchange information using protocols like OSPF and BGP.
Answer: Dynamic Routing
19. A routing protocol that uses "hop count" to find the shortest path.
Answer: RIP
20. A network security device that monitors and controls traffic based on predetermined rules.
Answer: Firewall
21. A firewall that operates at Layer 3 and makes decisions based on packet headers.
Answer: Packet Filtering Firewall
22. A firewall that tracks active connections to make intelligent filtering decisions.
Answer: Stateful Inspection Firewall
23. A firewall that operates at Layer 7 and understands specific protocols like HTTP or FTP.
Answer: Application-Layer Firewall
24. The user-friendly command-line interface for managing Linux firewall rules.
Answer: UFW
25. The command to enable the Uncomplicated Firewall.
Answer: `sudo ufw enable`
26. The command to check the status of the Uncomplicated Firewall.
Answer: `sudo ufw status`
27. The tool used to capture and analyze network data in real time.
Answer: Wireshark
28. A tool used for network discovery, identifying open ports, and security auditing.
Answer: Nmap
29. An open-source intrusion detection and prevention system (IDS/IPS).
Answer: Snort

30. A method used by firewalls to translate private IP addresses into public IP addresses.
Answer: NAT
31. A security strategy that employs multiple layers of defense mechanisms.
Answer: Defense in Depth
32. A security principle that restricts users' access rights to only what is necessary for their role.
Answer: Least Privilege
33. The file path for the main Apache2 configuration.
Answer: `/etc/apache2/`
34. The file used by Apache to determine listening ports.
Answer: `ports.conf`
35. The command used to update the system package list.
Answer: `sudo apt update`
36. The command to install the OpenSSH server.
Answer: `sudo apt install openssh-server`
37. The protocol that specifies the origin and destination for each data packet.
Answer: IP (Internet Protocol)
38. The autonomous system routing protocol used to announce which networks control which IP addresses.
Answer: BGP
39. The term for when a packet moves from one network to another.
Answer: Hop
40. The number of steps generally associated with firewall testing in your materials.
Answer: 13
-

Part 2: Multiple Choice

1. Which command is used to restart the SSH service?
Answer: b) `sudo systemctl restart ssh`
2. What does the "DHCP Offer" message do in the DHCP process?
Answer: c) Provides IP address lease information to the client

3. Which OSI layer does a "Packet Filtering Firewall" typically operate at?
Answer: b) Layer 3
4. What is the command to allow traffic for port 80/tcp in UFW?
Answer: c) `sudo ufw allow 80/tcp`
5. Which protocol is commonly used to dynamically identify the fastest and shortest paths within an Autonomous System?
Answer: c) OSPF
6. What is the primary role of a firewall?
Answer: c) To monitor and control traffic based on security rules
7. Which of the following is NOT a layer of the TCP/IP stack?
Answer: b) Session Layer
8. If you want to view the current status of UFW rules, which command should you use?
Answer: c) `sudo ufw status`
9. What information does a "Stateful Inspection Firewall" maintain to make filtering decisions?
Answer: b) A state table tracking active connections
10. Which tool is best suited for identifying open ports and running services on a network?
Answer: c) Nmap

Quick Skimming

I. Basics of Computer Networks

A **computer network** is a system that connects multiple devices (such as computers, printers, and servers) so they can communicate and share resources. These resources can include files, applications, and internet access.

Without networks, devices would work independently, and transferring data would require physical storage devices like USB drives.

Types of Networks:

- **Local Area Network (LAN)**
A LAN connects devices within a small area like a home, school, or office. Because devices are close together, communication is fast and usually managed by routers or switches.
- **Wide Area Network (WAN)**
A WAN connects devices across large distances, even globally. It links multiple LANs together. The best example is the internet.

II. OSI Model (7 Layers Explained)

The **OSI Model** is a framework that explains how data travels through a network in seven layers. Each layer has a specific function.

Layers (Top to Bottom):

- **Application Layer**
This is where users interact with the network. It provides services for applications like web browsers and email.
- **Presentation Layer**
This layer formats and translates data. It also handles encryption to secure data.
- **Session Layer**
It manages communication sessions

between devices, including starting and ending connections.

- **Transport Layer**
Ensures data is delivered correctly and in order. It also handles error detection and recovery (e.g., TCP).
- **Network Layer**
Responsible for routing data using IP addresses to reach the correct destination.
- **Data Link Layer**
Handles communication between devices on the same network using MAC addresses.
- **Physical Layer**
Deals with the physical transmission of data (cables, signals, hardware).

👉 The OSI model helps troubleshoot network problems by identifying which layer is causing the issue.

III. TCP/IP Stack

The **TCP/IP model** is the practical version of networking used on the internet.

Layers:

- **Application Layer** – Handles user-level protocols (like HTTP)
- **Transport Layer** – Ensures reliable communication
- **Internet Layer** – Handles IP addressing and routing
- **Network Access Layer** – Manages physical transmission

Key Protocols:

- **TCP (Transmission Control Protocol)**
Ensures data is delivered accurately and in the correct order.
- **IP (Internet Protocol)**
Identifies where data is coming from and where it should go.

IV. Common Network Issues (With Explanation)

- **Connection Issues**
Occur when devices cannot connect to a network or disconnect frequently.
Example: A laptop cannot connect to Wi-Fi.
- **Slow Network Performance**
Happens when data transfers slowly due to congestion or weak signals.
Example: Websites take too long to load.
- **Signal Interference Problems**
Wireless signals become weak or unstable due to obstacles or interference.
Example: Wi-Fi drops in certain rooms.
- **DHCP Configuration Issues**
Errors in automatic IP assignment can cause conflicts.
Example: Two devices get the same IP address.
- **DNS Resolution Problems**
The system cannot convert domain names into IP addresses.
Example: A website shows a DNS error.
- **Firewall Blocking**
Access is restricted due to security rules.
Example: A website or service is blocked.
- **Routing Issues**
Data is sent through incorrect paths.
Example: Devices on different networks cannot communicate.
- **Hardware Failures**
Network devices malfunction.
Example: Router keeps restarting.
- **Security Threats**
Attacks such as malware or phishing compromise the network.
- **Bandwidth Saturation**
Too many users consume network resources, slowing it down.

V. Routing

Routing is the process of finding the best path for data to travel across networks.

Routing Process:

- **Destination Address** – Identifies where the data should go
- **Routing Table** – Stores possible paths
- **Routing Decision** – Chooses the best path
- **Forwarding** – Sends data to the next device

Types:

- **Static Routing** – Manually configured routes
- **Dynamic Routing** – Automatically updated routes

Routing Protocols:

- **BGP** – Used for large-scale networks (like the internet)
- **OSPF** – Finds the shortest path efficiently
- **RIP** – Uses hop count to determine routes

VI. Firewall

A **firewall** is a security system that controls network traffic based on rules.

It acts as a barrier between:

- Trusted internal networks
- Untrusted external networks

Its main purpose is to prevent unauthorized access and protect against attacks.

VII. Types of Firewalls

- **Packet Filtering Firewall**
Checks basic packet information like IP address and port.
- **Stateful Inspection Firewall**
Monitors active connections and makes decisions based on traffic state.
- **Application-Layer Firewall**
Inspects traffic deeply at the application level (more secure).

VIII. Firewall Configuration Example

A firewall can be configured to control specific traffic:

- Allow web traffic (HTTP) on port 80
- Block Telnet traffic on port 23

👉 This shows how firewalls allow safe services while blocking risky ones.

IX. Firewall Vulnerabilities and Threats

- **DDoS Attacks** – Overwhelm network with traffic
- **Insider Attacks** – Threats from within the organization
- **Outdated Software** – Security holes due to lack of updates
- **Improper Configuration** – Weak or incorrect firewall rules

X. UFW (Uncomplicated Firewall)

UFW is a simple firewall tool used in Linux.

Commands Explained:

- `sudo ufw enable` – Turns the firewall on
- `sudo ufw disable` – Turns it off
- `sudo ufw status` – Shows current rules
- `sudo ufw allow ssh` – Allows SSH access
- `sudo ufw allow 80/tcp` – Allows web traffic
- `sudo ufw deny 23/tcp` – Blocks Telnet
- `sudo ufw delete <rule_number>` – Removes a rule

👉 The firewall must be enabled for rules to take effect.

XI. Network Security Tools

- **Wireshark** – Captures and analyzes network packets
- **Nmap** – Scans networks for hosts and open ports
- **Snort** – Detects and prevents intrusions

XII. Network Security Best Practices

- **Defense in Depth**
Use multiple layers of security instead of relying on one.
- **Least Privilege**
Users only get the access they truly need.
- **Security Policies**
Clear rules for managing and protecting systems.

LINUX SERVICES AND DHCP

XIII. Services and Daemons

- A **service** is a program that runs in the background to provide functionality.
- A **daemon** is a type of service that starts automatically and runs continuously.

Examples:

- `sshd` – Handles remote login (SSH)
- `dhcpd` – Manages IP address assignment

XIV. Managing Services (systemctl)

Used to control services in Linux:

- Start → `sudo systemctl start ssh`
- Stop → `sudo systemctl stop ssh`
- Restart → `sudo systemctl restart ssh`
- Enable → `sudo systemctl enable ssh`
- Status → `sudo systemctl status ssh`

👉 Restarting ensures new configurations are applied.

XV. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns network settings like:

- IP address
- Subnet mask
- Default gateway
- DNS server

👉 Without DHCP, these would need to be set manually.

XVI. DHCP Components

- **DHCP Server** – Assigns IP addresses
- **DHCP Client** – Receives configuration
- **DHCP Relay Agent** – Passes requests between networks

XVII. DHCP Process (Step-by-Step)

1. **Discover** – Client searches for DHCP server
2. **Offer** – Server offers an IP address
3. **Request** – Client accepts the offer
4. **Acknowledgment** – Server confirms assignment

XVIII. DHCP Concepts

- **Lease** – Duration an IP is assigned
- **Scope** – Range of IP addresses available
- **Reservation** – Assigning a fixed IP to a specific device

XIX. DHCP Server Configuration (Linux)

Steps:

1. Set a static IP

2. Install DHCP server
3. Edit configuration file
4. Define IP range and options
5. Start and test service

Example:

range 192.168.10.50 192.168.10.100;
option routers 192.168.10.1;

XX. Web Server Setup (Apache)

Steps:

1. Install Apache
2. Start the service
3. Enable it at startup
4. Allow HTTP in firewall
5. Test in browser

👉 Website files are stored in `/var/www/html`.

XXI. File Sharing (Samba)

Samba allows file sharing between Linux and Windows systems.

Steps:

1. Create shared folder
2. Edit configuration file
3. Restart service
4. Access from another device

XXII. FTP and SFTP

- **FTP** – Transfers files but is not secure
- **SFTP** – Secure file transfer using SSH

Commands:

- Upload → `put file.txt`
- Download → `get file.txt`

XXIII. Squid Proxy Server

A **proxy server** acts as a middleman between users and the internet.

It can:

- Control access to websites
- Monitor usage
- Log activity

Configured via:

`/etc/squid/squid.conf`

XXIV. Ports

Ports identify services running on a device.

Examples:

- FTP → Port 21
- SSH → Port 22

👉 Ports allow multiple services to run on the same device.

XXV. Multi-Service Server

One server can run multiple services at the same time:

- **Samba** – File sharing
- **FTP (vsftpd)** – File transfer
- **SFTP** – Secure transfer
- **Squid** – Web proxy

👉 Each service must be configured and tested properly to avoid conflicts.

Comprehensive notes

NETWORK AND FIREWALL LINUX SERVICES

PART 1: BASICS OF COMPUTER NETWORKS

Computer networks are systems designed to connect multiple devices together so they can **communicate** and **share resources**. These devices include computers, servers, printers, and other peripherals, which can exchange data and share things like files, applications, and internet access.

Think of a network like a postal system — each device has an address, and messages (data) are delivered from one address to another. The network provides the roads, the devices are the houses, and the data packets are the letters.

Types of Networks

- **LAN (Local Area Network):** A network that covers a small geographic area, such as a home, office, or school building. Devices on a LAN are usually connected via Ethernet cables or Wi-Fi.
- **WAN (Wide Area Network):** A network that spans large geographic areas, such as cities or countries. The internet itself is the largest WAN in existence.

PART 2: THE OSI MODEL

The **OSI Model (Open Systems Interconnection Model)** is a conceptual framework that standardizes the functions of a telecommunication or computing system into **seven abstraction layers**. It was created to help different devices and systems communicate with one another by following a common set of rules. These layers help in understanding and designing network communication protocols and interactions.

The seven layers, from top (closest to the user) to bottom (closest to hardware), are:

Layer #	Layer Name	What It Does
7	Application Layer	The layer closest to the user. It provides network services directly to applications (e.g., web browser, email).
6	Presentation Layer	Translates data into a format the application layer can accept. Handles encryption and data compression.
5	Session Layer	Manages sessions (connections) between applications. Establishes, maintains, and terminates connections.
4	Transport Layer	Provides reliable data transfer between hosts. Uses TCP (reliable) or UDP (faster but less reliable).
3	Network Layer	Handles routing of data packets across networks. This is where IP addresses are used. Firewalls often operate here.

2	Data Link Layer	Handles communication between adjacent devices on the same network. Uses MAC addresses.
1	Physical Layer	The actual physical hardware — cables, switches, electrical signals, radio waves.

Memory trick: **"All People Seem To Need Data Processing"** (Application, Presentation, Session, Transport, Network, Data Link, Physical) — read from top to bottom.

PART 3: TCP/IP STACK

The **TCP/IP Stack (Transmission Control Protocol / Internet Protocol)** is the actual suite of protocols used for communication over the internet and most local networks today. It is named after its two most important protocols: **TCP** and **IP**. Unlike the OSI Model (which has 7 layers), the TCP/IP Stack has only **four layers**.

TCP/IP Layer	Corresponding OSI Layers	Function
Application Layer	OSI 5, 6, 7 (Session, Presentation, Application)	Handles high-level protocols like HTTP, FTP, SMTP, DNS.
Transport Layer	OSI 4 (Transport)	Reliable data delivery using TCP; fast delivery using UDP.
Internet Layer	OSI 3 (Network)	IP addressing and routing of packets across networks.
Network Access Layer	OSI 1, 2 (Physical, Data Link)	Physical transmission of data — Ethernet, Wi-Fi, MAC.

PART 4: ROUTING AND ROUTING PROTOCOLS

Routing is the process of selecting the best path for network traffic to travel from its source to its destination across one or more interconnected networks. **Routers** are the devices that perform routing — they forward data packets between networks based on the packet's destination **IP address**.

The Concept of Routing

When data travels across a network, here is what happens step by step:

- **1. Destination Address:** When a device wants to send data to another device on a different network, it uses the destination IP address to figure out where the data needs to go.
- **2. Routing Table:** Each router maintains a routing table — think of it as a GPS map. It is a list of known networks and the next-hop routers (the next stop) through which packets should be forwarded to reach those networks.

- **3. Routing Decision:** When a router receives a packet, it looks at the destination IP address and consults its routing table to determine the best path to forward the packet.
- **4. Forwarding:** Once the router determines the next hop, it forwards the packet along that path until it reaches the destination.

Types of Routing

- **Static Routing:** In static routing, network administrators manually configure the routing table on each router. This works well for small, simple networks or for routes that rarely change. It is predictable but requires manual updates when the network changes.
- **Dynamic Routing:** In dynamic routing, routers automatically exchange routing information with each other using routing protocols. Dynamic routing protocols such as **OSPF (Open Shortest Path First)** and **BGP (Border Gateway Protocol)** automatically update routing tables when the network changes. Dynamic routing is more scalable and flexible than static routing, but it requires more configuration and has more management overhead.

Main Routing Protocols

A **protocol** is a standardized way of formatting data so that any connected computer can understand it. A **routing protocol** is specifically used for identifying or announcing network paths. The main routing protocols are:

- **IP (Internet Protocol):** Specifies the origin and destination for each data packet. Routers inspect each packet's IP header to identify where to send them.
- **BGP (Border Gateway Protocol):** Used to announce which networks control which IP addresses and which networks connect to each other. The large networks that make BGP announcements are called **autonomous systems (AS)**. BGP is a **dynamic routing protocol** used between large networks on the internet.
- **OSPF (Open Shortest Path First):** Commonly used by network routers within a single organization (within an AS) to dynamically identify the fastest and shortest available routes for sending packets to their destination.
- **RIP (Routing Information Protocol):** Uses "**hop count**" to find the shortest path from one network to another. Hop count means the number of routers a packet must pass through. When a packet goes from one network to another, this is called a "**hop**".

PART 5: COMMON NETWORK ISSUES

Network problems happen in real-world environments. Here are the ten most common issues and what causes them:

1. **Connection Issues:** Devices cannot connect to the network or experience intermittent connectivity (keeps dropping). Example: A laptop unable to connect to the Wi-Fi network despite being in range.
2. **Slow Network Performance:** Slow data transfer speeds or delays in accessing network resources. Example: Websites loading slowly or files taking a long time to download.
3. **Interference or Signal Strength Problems:** Wireless networks experience interference or low signal strength. Example: Wi-Fi signal dropping or becoming weak in certain areas of a building.
4. **DHCP Configuration Issues:** Incorrect DHCP settings can lead to IP address conflicts or devices unable to obtain IP addresses. Example: Multiple devices assigned the same IP address due to DHCP misconfiguration.
5. **DNS Resolution Problems:** The network is unable to resolve domain names (like google.com) to IP addresses, causing websites not to load. Example: Typing a website address in the browser but receiving a "DNS error" message.
6. **Firewall Blocking:** Firewall rules blocking certain network traffic, leading to restricted access to resources. Example: Unable to access a specific website or service due to firewall restrictions.
7. **Routing Issues:** Incorrect routing configuration causing packets to be sent to the wrong destination. Example: Unable to communicate with devices on a different subnet within the same network.
8. **Hardware Failures:** Physical network hardware failures such as faulty routers, switches, or network interface cards (NICs). Example: Router continuously rebooting or network switch ports not functioning properly.
9. **Security Threats:** Network security breaches, malware infections, or unauthorized access attempts. Example: Data breach due to a phishing attack or malware spreading across the network.
10. **Bandwidth Saturation:** The network becomes overloaded with traffic, causing congestion and performance slowdowns. Example: Slow network speeds during peak hours when many users are accessing the network simultaneously.

PART 6: FIREWALLS

A **firewall** is a network security device or software that **monitors and controls incoming and outgoing network traffic** based on predetermined security rules. Its primary role is to establish a **barrier** between a trusted internal network and untrusted external networks (such as the internet), thereby protecting the internal network from unauthorized access and malicious activities.

Think of a firewall like a security guard at a building entrance. The guard checks everyone going in and out. If you are on the approved list (firewall rule allows it), you can pass. If not, you are denied entry.

Primary Function of a Firewall

The primary function of a firewall is to **filter packets** — inspect data packets and allow or deny them based on rules. A firewall rule allowing only HTTP traffic while denying all other inbound connections is mainly designed to **limit network exposure to only a specific service**.

Types of Firewalls (by filtering method)

1. **Packet Filtering Firewall:** Operates at the **Network Layer (Layer 3)** of the OSI model. It examines individual packets of data as they pass through the firewall and makes decisions based on predefined rules. Rules are typically based on information in the packet headers, such as source and destination IP addresses, port numbers, and protocol types. Packet filtering firewalls are simple and efficient, but they **cannot inspect the contents of packets** beyond header information.
2. **Stateful Inspection Firewall:** Operates at the **Network and Transport Layers (Layers 3 and 4)** of the OSI model. In addition to examining packet headers, it keeps track of the **state of active connections** (e.g., TCP sessions) to make more intelligent filtering decisions. Stateful inspection firewalls maintain a **state table** that tracks the state of each connection, allowing them to enforce more granular security policies. They provide better security and performance compared to packet filtering firewalls.
3. **Application-Layer Firewall (Proxy Firewall):** Operates at the **Application Layer (Layer 7)** of the OSI model. It inspects and filters network traffic at the application level, which allows it to understand and control specific application protocols (e.g., HTTP, FTP, SMTP). Application-layer firewalls can enforce detailed security policies based on application-specific criteria, such as URL filtering, content inspection, and user authentication. They provide the **highest level of security** but may introduce additional latency due to deep packet inspection.

Firewall Deployment Methods

Firewalls can be classified based on their deployment methods into three types:

- **Hardware Firewalls:** Physical devices placed between the network and gateway.
- **Software Firewalls:** Programs installed on a computer to filter traffic for that device.
- **Cloud Firewalls:** Firewall services delivered over the cloud, protecting cloud infrastructure.

Firewall Testing — 13 Steps

To verify a firewall is working correctly, the following 13 steps are performed:

1. Locating the firewall
2. Running traceroute

3. Scanning ports
4. Banner grabbing
5. Access control enumeration
6. Identifying the firewall architecture
7. Testing the firewall policy
8. Firewalking

9. Port redirection
10. Internal and external testing
11. Testing for covert channels
12. HTTP tunneling
13. Identifying firewall-specific vulnerabilities

Top 7 Firewall Vulnerabilities and Threats

Even firewalls themselves can be vulnerable if not properly managed. The top 7 threats are:

1. **DDoS Attacks:** Attackers flood the firewall or network with so much traffic it can no longer function.
2. **Insider Attacks:** Threats that come from within the organization — employees who misuse their access.
3. **Outdated Firewall Software:** Old firmware or software that has unpatched security vulnerabilities.
4. **Failure to Activate Controls:** The firewall is installed but not properly enabled or configured.
5. **Lack of Documentation:** No records of what rules exist and why, leading to poor management.
6. **Basic Inspection Protocols:** Using only simple packet filtering when deeper inspection is needed.
7. **Improper Configuration:** Misconfigured rules that either block legitimate traffic or allow dangerous traffic.

Example: Packet Filtering Firewall Rule

Consider a packet filtering firewall that needs to allow HTTP (port 80) traffic while blocking Telnet (port 23) traffic. The firewall administrator defines the following rules:

- **Allow HTTP (Port 80) Traffic:** Rule: Allow inbound TCP traffic with destination port 80. Action: Permit the packet. This means any device can reach the web server.
- **Block Telnet (Port 23) Traffic:** Rule: Block inbound TCP traffic with destination port 23. Action: Drop or reject the packet. This prevents insecure remote connections.

PART 7: UFW (UNCOMPLICATED FIREWALL)

UFW (Uncomplicated Firewall) is a user-friendly command-line interface for managing **netfilter firewall rules** in Linux distributions. It provides a simplified way to configure and manage firewall rules without needing to deal with complex iptables commands directly. UFW is designed to make firewall management more accessible for users who may not be familiar with iptables syntax.

Basic UFW Commands and Syntax

Here are the essential UFW commands you need to know:

Command	What It Does
<code>sudo ufw enable</code>	Activates/enables the UFW firewall
<code>sudo ufw disable</code>	Deactivates/disables the UFW firewall
<code>sudo ufw status</code>	Displays the current status and active rules of UFW
<code>sudo ufw allow ssh</code>	Allows SSH traffic (port 22) through the firewall
<code>sudo ufw allow 80/tcp</code>	Allows HTTP traffic on port 80 (TCP)
<code>sudo ufw allow samba</code>	Allows Samba file-sharing traffic through the firewall
<code>sudo ufw deny ftp</code>	Denies FTP traffic through the firewall
<code>sudo ufw deny 23/tcp</code>	Blocks Telnet traffic on port 23
<code>sudo ufw delete <rule_number></code>	Deletes a specific firewall rule by its number

Important Note: After configuring firewall rules using UFW, it is essential to enable the firewall to activate the rules. UFW automatically starts at boot once enabled. Always verify rules using **sudo ufw status** to ensure they are applied correctly.

PART 8: NETWORK SECURITY TOOLS

Network administrators use various tools to monitor, analyze, and secure their networks. The three major tools are:

1. **Wireshark:** Wireshark is a widely used open-source **network protocol analyzer**. It allows users to capture and analyze the data traveling back and forth on a network in real-time — inspecting actual packet content. Use Cases: Troubleshooting network issues, analyzing network protocols, detecting security threats by inspecting packet-level details.
2. **Nmap (Network Mapper):** Nmap is a powerful open-source tool for **network discovery and security auditing**. It can discover hosts and services on a computer network, identifying open ports, running services, and their version details. Use Cases: Network mapping, vulnerability assessment, identifying potential security risks.
3. **Snort:** Snort is a widely used open-source **intrusion detection and prevention system (IDS/IPS)**. It analyzes network traffic in real-time and can generate alerts or take actions based on predefined rulesets. Use Cases: Detecting and preventing malicious activities on a network, including intrusion attempts and malware.

PART 9: ADVANCED FIREWALL CONFIGURATION

- **NAT (Network Address Translation):** NAT is a method used by firewalls to **translate private IP addresses** used within a local network into **public IP addresses** used on the internet. It allows multiple devices within a private network to share a single public IP address for internet access. NAT also provides security by hiding internal IP addresses from external networks.

- **VPN Passthrough:** VPN passthrough allows VPN (Virtual Private Network) traffic to pass through the firewall without being blocked. It is used to enable VPN connections initiated from within the internal network to external VPN servers. This is essential for organizations that use VPN technology to securely connect remote users or branch offices to the corporate network.
- **Application-Layer Filtering:** This firewall feature inspects network traffic at the Application Layer (Layer 7 of the OSI model) to filter or block specific applications or protocols. Use Cases: Blocking unauthorized applications like peer-to-peer file sharing, instant messaging, or specific web applications.

PART 10: NETWORK SECURITY BEST PRACTICES

- **Defense in Depth:** Defense in depth is a security strategy that employs **multiple layers of defense mechanisms** to protect networks and systems. Instead of relying on a single security measure, it uses a combination of physical, technical, and administrative controls to mitigate risks and threats. In network security, this includes perimeter security (firewalls, intrusion detection systems), network segmentation, access controls (authentication, authorization), encryption, and monitoring.
- **Least Privilege:** Least privilege is a security principle that restricts users' access rights to **only what is necessary for their role or function**. It aims to minimize the potential impact of security breaches. In network security, this means granting users and devices only the necessary access rights and permissions to perform their intended tasks — limiting access to sensitive data, resources, and network services to authorized users only.
- **Security Policies:** Security policies are formal documents that outline an organization's guidelines, rules, and procedures for protecting its information assets. They define rules and requirements for access control, data protection, network monitoring, incident response, and other security-related activities. They help establish a baseline for security practices, ensure consistency in security enforcement, and promote accountability across the organization.

PART 11: LINUX SERVICES AND DAEMONS

What is a Service?

A **service** is a program that runs in the background to provide functionality to the system or to other programs. Services are always running, even when no user is logged in. You do not see them on your screen — they work silently behind the scenes.

Examples of services include:

- SSH server — allows remote login to the system
- Web server — serves web pages to clients
- DHCP server — automatically assigns IP addresses to devices

What is a Daemon?

A **daemon** is a **background process (service)** that starts automatically when the system boots and waits for requests. The word "daemon" comes from Greek mythology — background spirits that perform tasks without being seen. Daemon names in Linux typically end in the letter "**d**".

Examples:

- **sshd** → The SSH daemon (manages incoming SSH connections)
- **dhcpcd** → The DHCP daemon (manages IP address assignments)

In Ubuntu Server, services and daemons are managed by **systemd**, the system and service manager.

Managing Services with systemctl

Ubuntu uses **systemctl** to manage services. Here are the most important commands:

Command	What It Does
<code>sudo systemctl start ssh</code>	Starts the SSH service immediately
<code>sudo systemctl stop ssh</code>	Stops the SSH service
<code>sudo systemctl restart ssh</code>	Stops then starts the service again — applies config changes
<code>sudo systemctl reload ssh</code>	Reloads config WITHOUT fully restarting (not the same as restart)
<code>sudo systemctl enable ssh</code>	Makes service start automatically at boot
<code>sudo systemctl status ssh</code>	Checks and displays the current status of the service

Important distinction — restart vs reload: The correct command to restart a service and apply updated configuration is **sudo systemctl restart <service>**, not reload. For example, to restart Squid: **sudo systemctl restart squid**. Restarting a service ensures that updated configurations are applied.

PART 12: DHCP (Dynamic Host Configuration Protocol)

DHCP (Dynamic Host Configuration Protocol) is a network protocol that **automatically assigns network configuration information to devices** on a network. Without DHCP, every computer must be configured manually — which is impractical at scale.

DHCP automatically assigns the following to each device:

- **IP Address** — the unique address for the device on the network
- **Subnet Mask** — defines the network portion of the IP address
- **Gateway** — the router address the device uses to reach other networks
- **DNS Server** — the server that resolves domain names to IP addresses

DHCP Components

- **DHCP Server:** Responsible for leasing IP addresses and other network configuration parameters to DHCP clients. In Linux, this is the **isc-dhcp-server** package.
- **DHCP Client:** Devices that request and receive network configuration information from DHCP servers. When a client boots up, it asks: "Is there a DHCP server? I need an IP address!"
- **DHCP Relay Agent:** In networks with multiple subnets, a DHCP relay agent forwards DHCP messages between clients and servers across different network segments.

The DHCP Process (DORA)

The DHCP process has four steps, remembered by the acronym DORA:

- **D — DHCP Discover:** The client device broadcasts a DHCP discover message to the entire network to locate available DHCP servers. It is like shouting: "Is there a DHCP server out there?"
- **O — DHCP Offer:** DHCP servers respond with DHCP offer messages, providing IP address lease information to the client. The server says: "Yes, I'm here! Here is an IP address you can use."
- **R — DHCP Request:** The client selects one offer (if multiple servers respond) and sends a DHCP request message to formally request the offered IP address. The client replies: "Thank you, I would like to use that IP address."
- **A — DHCP Acknowledgement:** The selected DHCP server acknowledges the request and assigns the IP address to the client, completing the DHCP process. The server confirms: "It's yours! Use this address for the next [lease time]."

DHCP Lease

A **DHCP lease** is the duration for which an IP address is allocated to a client. It defines how long the client can use the assigned IP address. DHCP leases can be configured with specific durations, typically ranging from hours to days. Clients may renew their leases with the DHCP server before they expire to continue using the assigned IP address.

DHCP Scope

A **DHCP scope** is a range of IP addresses from which the DHCP server can assign addresses to clients. The DHCP server manages a pool of available IP addresses within a

scope, dynamically allocating them to clients as needed. Administrators can define the range of addresses in a scope, as well as **exclude specific addresses** from being assigned (for example, reserving addresses for printers or servers).

DHCP Reservation

DHCP reservations allow administrators to assign **specific IP addresses to specific devices** based on their **MAC addresses**, ensuring that devices receive consistent IP assignments every time they connect. DHCP reservations are useful for devices that require static-like IP addresses, such as servers or network printers.

Installing and Configuring DHCP Server on Ubuntu

Key commands for DHCP server management:

Command	Purpose
<code>sudo apt install isc-dhcp-server</code>	Install the DHCP server package
<code>sudo nano /etc/dhcp/dhcpd.conf</code>	Edit the DHCP configuration file
<code>sudo systemctl restart isc-dhcp-server</code>	Restart the DHCP service to apply config changes
<code>sudo systemctl status isc-dhcp-server</code>	Check the status of the DHCP service
<code>sudo systemctl start isc-dhcp-server</code>	Start the DHCP service
<code>cat /var/lib/dhcp/dhcpd.leases</code>	View the current DHCP lease file

DHCP Configuration File: The correct path to the DHCP configuration file is `/etc/dhcp/dhcpd.conf`. This file is where you define the subnet, IP range (scope), gateway, DNS servers, and lease times.

DHCP Best Practices

- Regularly monitor DHCP server logs, lease status, and address usage to ensure smooth operation.
- Maintain accurate documentation of DHCP configurations, including scopes, reservations, options, and server settings.
- Deploy redundant DHCP servers and implement failover mechanisms to ensure high availability and reliability of DHCP services.

PART 13: NETWORK SERVICES — SAMBA, SQUID, FTP, SFTP

Overview of Services

Service	Purpose	Package Name
Samba	File sharing between Linux and Windows	samba (service: smbd)
Squid	Proxy server — web access control and caching	squid

FTP (vsftpd)	File transfer (unencrypted)	vsftpd
SFTP	Secure file transfer (encrypted, via SSH)	openssh-server

Samba — File Sharing (Linux and Windows)

The primary purpose of **Samba** in a network environment is to enable **file and printer sharing between Linux and Windows systems**. Samba allows Linux servers to appear as Windows file servers, enabling cross-platform file sharing using the SMB protocol. It is used for cross-platform file sharing.

Samba Setup Commands:

```
sudo apt install samba           # Install Samba
mkdir companysharedfile         # Create shared directory
sudo nano /etc/samba/smb.conf   # Edit Samba configuration
sudo systemctl restart smbd     # Restart Samba service
sudo ufw allow samba            # Allow Samba through
firewall
```

The **correct Samba service name** is **smbd**. The configuration file is located at **/etc/samba/smb.conf**.

In the Samba configuration, a shared folder entry looks like this:

```
[ShareFolder]
path = /share
browseable = yes
read only = no
guest ok = yes
```

Squid — Proxy Server

A **proxy server** is an intermediary server that processes client requests on their behalf. **Squid** is a widely-used proxy server in Linux used for **web caching and controlling web access**. Organizations use a proxy server to **monitor, control, and cache web access**. A company monitors and controls internet usage through an intermediary server that processes client requests — this describes what Squid does.

Squid Setup Commands:

```
sudo apt install squid           # Install Squid
sudo nano /etc/squid/squid.conf  # Edit configuration (correct
path)
sudo systemctl restart squid     # Restart Squid service
sudo tail /var/log/squid/access.log # View proxy access logs
```

Important: The correct Squid configuration file path is **/etc/squid/squid.conf**. The default Squid proxy port is **3128**.

FTP (File Transfer Protocol)

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and a server. FTP is **unencrypted** — this means credentials (username and password) and data are transmitted in plain text, making it vulnerable to interception.

FTP uses **port 21** for the control connection.

FTP Setup Commands (using vsftpd):

```
sudo apt install vsftpd           # Install FTP server
sudo nano /etc/vsftpd.conf         # Edit FTP configuration
sudo service vsftpd start         # Start FTP service (correct
command)
sudo systemctl restart vsftpd     # Restart FTP service
```

The **correct FTP package name** is **vsftpd**. The correct FTP service name is also **vsftpd**.

SFTP (Secure File Transfer Protocol)

SFTP (Secure File Transfer Protocol) is a **secure version of FTP**. The main distinction between FTP and SFTP is **the presence of encryption during authentication and data transfer**. SFTP operates as part of the **SSH (Secure Shell) protocol** — it is NOT an extension of FTP, but a completely separate protocol that uses SSH for security.

Key SFTP commands:

Command	What It Does
sftp username@192.168.1.10	Connect to an SFTP server securely
get report.pdf	Download (get) a file from the server — NOT "download"
put file.txt	Upload (put) a file to the server
ls	List files in the current directory

Important: An administrator uses a secure method of file transfer that protects credentials during transmission — this describes **SFTP**. The command to download a file in SFTP is **get <filename>**, not "download".

Ports and Why They Matter

A **port** is a numerical label used to identify a specific process or service on a network device. Ports are important because they allow **multiple services to run on one host simultaneously**. Think of an IP address as a building address, and a port as the specific apartment number inside that building.

Service	Port Number	Notes
FTP (Control)	21	Standard FTP control connection port
SSH / SFTP	22	Secure Shell and SFTP both use port 22
HTTP	80	Standard unencrypted web traffic

HTTPS	443	Encrypted web traffic
Squid Proxy	3128	Default Squid proxy port

To allow SSH connections through UFW, the correct command is **sudo ufw allow ssh** — NOT **sudo ufw allow 21** (which would allow FTP, not SSH).

PART 14: IP ADDRESSES

An **IP (Internet Protocol) address** is the numerical label assigned to each device in a network. It serves two main functions: identifying the host or network interface, and providing the location of the host in the network. IP addresses are how routers know where to send data packets.

DHCP stands for **Dynamic Host Configuration Protocol**. The word "**configuration**" is the full expansion of the letter C in DHCP.

PART 15: SETTING UP A WEB SERVER (Apache)

Setting up a web server on Ubuntu involves the following steps:

Step 1 — Set Up the VM: Create a Virtual Machine, attach Ubuntu ISO, and install with OpenSSH Server enabled.

Step 2 — Install and Enable SSH: Install openssh-server, enable and start it using systemctl.

Step 3 — Install Apache: Run `sudo apt install apache2 -y`

Step 4 — Check and Enable Apache: Verify with `sudo systemctl status apache2`; enable at boot with `sudo systemctl enable apache2`

Step 5 — Configure Firewall: Allow Apache through UFW: `sudo ufw allow 'Apache'`, then `sudo ufw enable`

Step 6 — Test: Navigate to `http://your-server-ip` in a browser and you should see the Apache2 Ubuntu Default Page

Step 7 — Custom Page: Navigate to `/var/www/html` and edit `index.html` with `sudo nano index.html`

QUICK REFERENCE SUMMARY

Term / Concept	Definition / Key Point
Firewall	A network security device/software that filters incoming and outgoing traffic based on rules.
DHCP	Dynamic Host Configuration Protocol — automatically assigns IP addresses and network config.
Daemon	A background process/service in Linux that starts automatically and waits for requests.
Proxy Server (Squid)	An intermediary server that monitors, controls, and caches web access.
Samba	Used for file and printer sharing between Linux and Windows systems.
FTP	Unencrypted file transfer protocol. Uses port 21. Package: vsftpd.
SFTP	Secure file transfer — part of SSH protocol. Encrypts credentials and data.
UFW	Uncomplicated Firewall — user-friendly interface for managing Linux firewall rules.
IP Address	Numerical label assigned to each device in a network.
Port 21	Standard FTP control connection port.
Port 22	SSH and SFTP port.
Port 80	HTTP (web) traffic port.
/etc/dhcp/dhcpd.conf	Correct DHCP configuration file path.
/etc/squid/squid.conf	Correct Squid configuration file path.
/etc/samba/smb.conf	Samba configuration file.
smbd	Correct Samba service name used with systemctl.
vsftpd	Correct FTP package and service name.
DORA Process	DHCP process: Discover, Offer, Request, Acknowledgement.
Defense in Depth	Using multiple layers of security controls to protect networks.
Least Privilege	Users are given only the access they absolutely need for their role.
NAT	Network Address Translation — translates private IPs to public IPs.
Static Routing	Manually configured routing tables — simple but requires manual updates.
Dynamic Routing	Routers auto-update tables using protocols like OSPF and BGP.

Identification Sample

Section A - Service & Tool Names (Items 1 -12)

1. This is the Linux package and service name used for setting up an FTP server.
2. This is the service name used with systemctl to restart the Samba file-sharing service.
3. This service acts as an intermediary server that processes web requests from clients, controls internet access, and caches web content.
4. This is the Linux service that enables file and printer sharing between Linux and Windows systems.
5. This is the package name installed on Ubuntu to set up a DHCP server.
6. This is the background process in Linux that starts automatically at boot and waits for requests.
7. SFTP is included and operates as part of this underlying technology or protocol.
8. This is the name of the user-friendly Linux command-line tool used to manage netfilter firewall rules without using complex iptables commands directly.
9. This open-source network protocol analyzer allows users to capture and analyze data traveling across a network in real-time.
10. This open-source intrusion detection and prevention system (IDS/IPS) analyzes network traffic and generates alerts based on predefined rulesets.
11. This open-source network discovery and security auditing tool identifies open ports, running services, and their version details on a network.

12. This is the name of the web server software installed on Ubuntu to serve web pages to clients.

Section B — Commands (Items 13–35)

13. Write the correct command to install the Samba service on Ubuntu.
14. Write the correct command to restart the Samba service using systemctl.
15. Write the correct command to install the FTP server package on Ubuntu.
16. Write the correct command to start the FTP (vsftpd) service.
17. Write the correct command to install the DHCP server on Ubuntu.
18. Write the correct command to restart the DHCP server service.
19. Write the correct command to check the status of the DHCP server service.
20. Write the correct command to start the DHCP server service.
21. Write the correct command to enable the UFW firewall.
22. Write the correct UFW command to allow SSH connections.
23. Write the correct UFW command to allow Samba file sharing through the firewall.
24. Write the correct command to display the current UFW firewall status and active rules.
25. Write the correct command to restart the Squid proxy service.

26. Write the correct SFTP command to connect to a server with IP address 192.168.1.10 using the username admin.

27. In SFTP, write the correct command to download a file named report.pdf from the server.

28. In SFTP, write the correct command to upload a file named debt.txt to the server.

29. Write the correct command to open and edit the Squid proxy configuration file.

30. Write the correct command to open and edit the DHCP configuration file.

31. Write the correct command to open and edit the Samba configuration file.

32. Write the correct command to check the DHCP lease file and see assigned IP addresses.

33. Write the correct command to restart the SSH service using systemctl.

34. Write the correct command to check the status of the SSH service.

35. Write the correct command to enable a service (e.g., SSH) to start automatically at system boot.

Section C — File Paths & Port Numbers (Items 36–42)

36. What is the correct full file path of the DHCP server configuration file?

37. What is the correct full file path of the Squid proxy configuration file?

38. What is the correct full file path of the Samba configuration file?

39. What is the correct full file path of the FTP (vsftpd) configuration file?

40. What port number is used for standard FTP control connections?

41. What is the default port number used by the Squid proxy server?

42. What port number is shared by both SSH and SFTP?

ANSWER KEY

Network Services & Commands — Identification

No.	Question Summary	Answer
1	This is the Linux package and service name used for setting up an FTP server.	vsftpd
2	This is the service name used with systemctl to restart the Samba file-sharin...	smbd
3	This service acts as an intermediary server that processes web requests from ...	Squid
4	This is the Linux service that enables file and printer sharing between Linux...	Samba
5	This is the package name installed on Ubuntu to set up a DHCP server.	isc-dhcp-server
6	This is the background process in Linux that starts automatically at boot and...	daemon
7	SFTP is included and operates as part of this underlying technology or protocol.	SSH (Secure Shell protocol)
8	This is the name of the user-friendly Linux command-line tool used to manage ...	UFW (Uncomplicated Firewall)
9	This open-source network protocol analyzer allows users to capture and analyz...	Wireshark
10	This open-source intrusion detection and prevention system (IDS/IPS) analyzes...	Snort
11	This open-source network discovery and security auditing tool identifies open...	Nmap
12	This is the name of the web server software installed on Ubuntu to serve web ...	Apache (apache2)
13	Write the correct command to install the Samba service on Ubuntu.	sudo apt install samba
14	Write the correct command to restart the Samba service using systemctl.	sudo systemctl restart smbd
15	Write the correct command to install the FTP server package on Ubuntu.	sudo apt install vsftpd
16	Write the correct command to start the FTP (vsftpd) service.	sudo service vsftpd start
17	Write the correct command to install the DHCP server on Ubuntu.	sudo apt install isc-dhcp-server

18	Write the correct command to restart the DHCP server service.	<code>sudo systemctl restart isc-dhcp-server</code>
19	Write the correct command to check the status of the DHCP server service.	<code>sudo systemctl status isc-dhcp-server</code>
20	Write the correct command to start the DHCP server service.	<code>sudo systemctl start isc-dhcp-server</code>
21	Write the correct command to enable the UFW firewall.	<code>sudo ufw enable</code>
22	Write the correct UFW command to allow SSH connections.	<code>sudo ufw allow ssh</code>
23	Write the correct UFW command to allow Samba file sharing through the firewall.	<code>sudo ufw allow samba</code>
24	Write the correct command to display the current UFW firewall status and acti...	<code>sudo ufw status</code>
25	Write the correct command to restart the Squid proxy service.	<code>sudo systemctl restart squid</code>
26	Write the correct SFTP command to connect to a server with IP address 192.168...	<code>sftp admin@192.168.1.10</code>
27	In SFTP, write the correct command to download a file named report.pdf from t...	<code>get report.pdf</code>
28	In SFTP, write the correct command to upload a file named debt.txt to the ser...	<code>put debt.txt</code>
29	Write the correct command to open and edit the Squid proxy configuration file.	<code>sudo nano /etc/squid/squid.conf</code>
30	Write the correct command to open and edit the DHCP configuration file.	<code>sudo nano /etc/dhcp/dhcpd.conf</code>
31	Write the correct command to open and edit the Samba configuration file.	<code>sudo nano /etc/samba/smb.conf</code>
32	Write the correct command to check the DHCP lease file and see assigned IP ad...	<code>cat /var/lib/dhcp/dhcpd.leases</code>
33	Write the correct command to restart the SSH service using systemctl.	<code>sudo systemctl restart ssh</code>
34	Write the correct command to check the status of the SSH service.	<code>sudo systemctl status ssh</code>
35	Write the correct command to enable a service (e.g., SSH) to start automatica...	<code>sudo systemctl enable ssh</code>
36	What is the correct full file path of the DHCP server configuration file?	<code>/etc/dhcp/dhcpd.conf</code>

37	What is the correct full file path of the Squid proxy configuration file?	/etc/squid/squid.conf
38	What is the correct full file path of the Samba configuration file?	/etc/samba/smb.conf
39	What is the correct full file path of the FTP (vsftpd) configuration file?	/etc/vsftpd.conf
40	What port number is used for standard FTP control connections?	Port 21
41	What is the default port number used by the Squid proxy server?	Port 3128
42	What port number is shared by both SSH and SFTP?	Port 22

EXAM RESULTS v2

1. Which statement most accurately describes the primary function of a firewall?

A. **Filters inbound and outbound packets according to defined security policies**

B. Automatically distributes IP configurations to network clients

C. Resolves domain names into numerical IP addresses

D. Maintains records of network traffic for auditing purposes

2. Which situation most clearly illustrates the operation of a firewall?

A. A domain name is translated into its corresponding IP address

B. A file is transferred between two computers using a secure protocol

C. **A server denies an incoming connection attempt on port 22 from an unknown source**

D. A workstation automatically receives a network configuration from a server

3. A firewall rule allowing only HTTP traffic while denying all other inbound connections is mainly designed to:

A. Automatically assign addresses to client machines

B. **Limit network exposure to only a specific service**

C. Prevent domain name resolution within the network

D. Improve overall network throughput

4. What is the purpose of DHCP?

A. Block traffic automatically

B. **Assign IP addresses automatically**

C. Share files automatically

D. Encrypt data automatically

5. Which file typically stores DHCP configurations?

A. /etc/network/interfaces

B.

/etc/systemd/system/isc-dhcp-server.service

C. /etc/default/isc-dhcp-server

D. **/etc/dhcp/dhcpd.conf**

6. What is the fundamental role of a proxy server?

A. Allocate network addresses dynamically

B. Encrypt communication channels between hosts

C. **Act as an intermediary that forwards client requests to external servers**

D. Authenticate users through directory services

7. Why do organizations commonly deploy a proxy server?

A. To replace DNS services in a local network

B. **To monitor, control, and cache web access within a network**

C. To automatically configure host IP addresses

D. To increase transmission latency for sensitive traffic

8. The software Squid is widely used for:

A. Secure remote file transfers between servers

B. Packet filtering at the network firewall level

C. Dynamic assignment of network configuration parameters

D. **Web request caching and access control for network clients**

9. What is the main distinction between File Transfer Protocol and Secure File Transfer Protocol?

A. **The presence of encryption during authentication and data transfer**

B. The operating system required for the server

C. The type of network topology required

D. The maximum file size allowed for transfer

10. The protocol SFTP operates as part of which underlying technology?

A. Domain Name System services

B. **Secure Shell protocol**

C. HTTP session tunneling

D. File Transfer Protocol extensions

11. Which port is typically used for standard FTP control connections?

A. **21**

B. 22

12. What is the primary purpose of Samba in a network environment?

- A. Filtering network packets between subnets
- B. Encrypting network traffic at the transport layer

C. **Enabling file and printer sharing between Linux and Windows systems**

- D. Automatically assigning IP addresses to network clients

13. Why are ports significant in computer networking?

- A. They define the maximum bandwidth available for a connection
- B. They encrypt packets during transmission across networks

C. **They allow multiple network services to operate simultaneously on a single host**

- D. They determine the physical interface used for data transmission

14. Restarting a system service after modifying its configuration primarily ensures that:

- A. Existing user sessions are terminated
- B. Temporary files created by the service are deleted

C. **Updated configuration settings are applied by the service**

- D. Network interface addresses are reassigned

15. What is a daemon?

A. **A background service that runs continuously to handle system or network requests**

- B. A kernel module responsible for hardware device communication

C. A command-line utility used to configure firewall rules

- D. A scheduled task executed at predefined intervals by the system scheduler

16. Which situation best describes a proxy server in action?

- A. Assigning IP addresses dynamically
- B. Filtering packets at kernel level

C. **Forwarding client requests to external servers**

- D. Encrypting file transfers automatically

17. While setting up a network, a device receives an IP address without manual configuration, indicating a DHCP service is active.

TRUE – DHCP automatically assigns IP addresses without manual configuration.

18. A technician observes that background processes start automatically during system boot and continue running without user interaction.

TRUE – Background services (daemons) start during boot and run continuously.

19. A company monitors and controls internet usage through an intermediary server that processes client requests.

TRUE – This describes a proxy server managing and monitoring requests.

20. An administrator uses a secure method of file transfer that protects credentials during transmission.

TRUE – Secure protocols like SFTP encrypt credentials during transfer.

21. A system administrator notices that incoming traffic is filtered before reaching internal services. He concludes that a firewall is likely active in the system.

TRUE – Firewalls filter incoming traffic before it reaches internal systems.

22. While configuring a network, an engineer observes that port numbers are used to identify specific services running on a host.

TRUE – Ports identify specific services running on a host.

23. An office shares files between Windows and Linux systems using a service that enables cross-platform file access.
TRUE – Samba enables file sharing between Linux and Windows systems.

24. A user transfers files over a network using a protocol that does not encrypt data.
TRUE – FTP does not encrypt data, making it insecure.

25. DHCP is responsible for assigning IP addresses automatically.
TRUE – DHCP's main function is automatic IP assignment.

26. SFTP provides encrypted file transfer over a network.
TRUE – SFTP encrypts both data and authentication.

27. A firewall allows all traffic by default unless configured.
FALSE – Firewalls do not allow all traffic by default; they enforce rules.

28. Proxy servers can cache web content to improve performance.
TRUE – Proxy servers cache data to improve speed and efficiency.

29. _____ is a secure version of FTP.
sftp

30. A _____ is used to control incoming and outgoing network traffic based on rules.
firewall

31. A proxy server like _____ controls web access.
squid

32. DHCP stands for Dynamic Host _____ Protocol.
configuration

33. The numerical label assigned to each device in a network is called an _____ address.
ip

34. _____ is used for file sharing between Linux and Windows.
samba

35. A background service in Linux is called a _____.
daemon

36. Restart Squid service
student@server:~\$ **sudo systemctl restart squid**

37. Download file using SFTP
sftp> **get report.pdf**

38. Squid configuration file path
/etc/squid/squid.conf

39. Allow SSH through firewall
student@server:~\$ **sudo ufw allow ssh**

40. Check DHCP service status
student@server:~\$ **sudo systemctl status isc-dhcp-server**

41. Start FTP service
student@server:~\$ **sudo systemctl start vsftpd**

42. Start DHCP service
student@server:~\$ **sudo systemctl start isc-dhcp-server**

43. DHCP configuration file
/etc/dhcp/dhcpd.conf

44. Secure connection to server

student@client:~\$ **sftp**
username@192.168.1.10

45. Upload file using SFTP

sftp> **put file.txt**

46. Allow Samba through firewall

student@server:~\$ **sudo ufw allow samba**

47. Correct FTP service name

vsftpd

48. Enable firewall

student@server:~\$ **sudo ufw enable**

49. Configure Firewall using UFW

Enable the firewall:

student@server:~\$ **sudo ufw enable**

Allow SSH connections:

student@server:~\$ **sudo ufw allow ssh**

Allow Samba file sharing:

student@server:~\$ **sudo ufw allow samba**

Display firewall status:

student@server:~\$ **sudo ufw status**

50. Setup Samba File Sharing

Install Samba:

student@server:~\$ **sudo apt install samba**

Create shared directory

(companysharedfile):

student@server:~\$ **mkdir**
companysharedfile

Edit Samba configuration file:

student@server:~\$ **sudo nano**
/etc/samba/smb.conf

Restart Samba service:

student@server:~\$ **sudo systemctl restart**
smbd

51. Configure DHCP Server

Install DHCP server:

student@server:~\$ **sudo apt install**
isc-dhcp-server

Edit DHCP configuration file:

student@server:~\$ **sudo nano**
/etc/dhcp/dhcpd.conf

Restart DHCP service:

student@server:~\$ **sudo systemctl restart**
isc-dhcp-server

Check DHCP status:

student@server:~\$ **sudo systemctl status**
isc-dhcp-server

52. Configure File Transfer Services

Install FTP server:

student@server:~\$ **sudo apt install vsftpd**

Connect using SFTP:

the username is hawakmoangbeat
the ip address is 192.168.1.10

student@client:~\$ **sftp**
hawakmoangbeat@192.168.1.10

Upload file using SFTP:

file name: debt.txt

sftp> **put debt.txt**

Semi Prep

Quiz 3

1. Fill in the Blank

Display disk usage in human-readable format

Answer: `df -h`

2. True or False

A system admin uses rsync daily to copy only new files. This is efficient

Answer: True

3. True or False

A disk is mounted, so users can access its files.

Answer: True

4. True or False

rsync copies only changed files.

Answer: True

5. Fill in the Blank

Mount `/dev/sdb1` to `/mnt/storage`

Answer: `sudo mount /dev/sdb1 /mnt/storage`

6. True or False

tar can compress files.

Answer: True

7. True or False

Full backup copies only modified files.

Answer: False

8. Multiple Choice

What does `df -h` display?

- A. Disk partitions
- B. Running processes
- C. File permissions
- D. Disk usage

Answer: Disk usage

9. True or False

umount is used to mount a disk.

Answer: False

10. Multiple Choice

What is the purpose of mounting?

- A. Delete files
- B. Make storage accessible
- C. Format disk
- D. Compress files

Answer: Make storage accessible

11. True or False

An admin edits `/etc/fstab` incorrectly and the system fails to mount drives. This can happen

Answer: True

12. Fill in the Blank

Extract `backup.tar`

Answer: `tar -xvf backup.tar`

13. Fill in the Blank

The command used to synchronize files is

Answer: `sudo rsync -av`

14. Fill in the Blank

Backup `/home/user` to `/backup` using `rsync`
Answer: `sudo rsync -av /home/user /backup`

15. Fill in the Blank

Unmount `/mnt/storage`
Answer: `sudo umount /mnt/storage`

16. Fill in the Blank

Delete a folder named `data`
Answer: `rm -r data`

17. Multiple Choice

Which command shows all disks and partitions?
A. `lsblk`
B. `df -h`
C. `mount`
D. `rsync`
Answer: `lsblk`

18. Fill in the Blank

Create a tar archive named `backup.tar` from folder `data`
Answer: `tar -cvf backup.tar data/`

19. Fill in the Blank

Show all disks and partitions
Answer: `lsblk`

20. Fill in the Blank

The command used to view disk structure is
Answer: `lsblk`

21. Fill in the Blank

Extract `backup.tar.gz`
Answer: `tar -xvzf backup.tar.gz`

22. Fill in the Blank

The command used to check disk space is
Answer: `df -h`

23. Multiple Choice

What does the `-z` option in `tar` do?
A. Extract
B. Rename
C. Delete
D. Compress
Answer: Compress

24. Fill in the Blank

The command used to create archive files is
Answer: `tar -cvf`

25. Fill in the Blank

Create a compressed tar archive `backup.tar.gz` from folder `data`
Answer: `tar -czvf backup.tar.gz data/`

26. True or False

A user deletes important files without backup and expects recovery. This is possible.

Answer: False

27. True or False

A system admin uses tar to combine many files into one archive. This is correct usage.

Answer: True

28. True or False

`/etc/fstab` is used for automatic mounting.

Answer: True

29. Fill in the Blank

The file used for automatic mounting is

Answer: `/etc/fstab`

30. Multiple Choice

Which tool is best for incremental backups?

- A. nano
- B. chmod
- C. rsync
- D. tar

Answer: rsync

Sample notes

* means “every value” in that field, - means a range, / means a step or interval, , means a list of exact values, L means last, W means nearest weekday, # means the Nth occurrence of a weekday in the month, and ? means “no specific value” in day fields.

1. `* /10 * * * *`
`/home/user/check.sh`
Runs every 10 minutes. `*/10` uses the slash step symbol.
2. `0 * * * *`
`/home/user/hourly.sh`
Runs once every hour, at minute 0. The * means every hour.
3. `30 2 * * *`
`/home/user/backup.sh`
Runs every day at 2:30 AM. Here `30` is the minute and `2` is the hour.
4. `0 9 * * 1-5`
`/home/user/workday-report.sh`
Runs Monday through Friday at 9:00 AM. `1-5` is a range for weekdays.
5. `0 8 * * 1,3,5`
`/home/user/cleanup.sh`
Runs at 8:00 AM on Monday, Wednesday, and Friday. The comma creates a list.
6. `0 0 1 * *`
`/home/user/monthly-report.sh`
Runs at midnight on the first day of every month.
7. `15 14 1 * *`
`/home/user/billing.sh`
Runs at 2:15 PM on the 1st of every month. This is a specific day-of-month job.
8. `0 0 1 1 *`
`/home/user/newyear.sh`
Runs at midnight on January 1 every year. This is the classic cron way to do a yearly task.

9. `0 0 * * 1L`
`/home/user/month-end.sh`
Runs on the last Monday of the month, using L.
10. `0 9 15W * *`
`/home/user/payroll.sh`
Runs on the weekday closest to the 15th. W means nearest weekday.

How the symbols work

* is the wildcard. It means “every possible value” for that field. Example: * in the hour field means every hour.

- is a range. Example: `9-17` in the hour field means from 9 AM to 5 PM.

/ is a step. Example: `*/10` in the minute field means every 10 minutes.

, is a list. Example: `1,3,5` in the day-of-week field means Monday, Wednesday, and Friday. This symbol appears in the table on your PDF.

L means last. In your PDF, `1L` in day-of-week means the last Monday of the month.

W means nearest weekday. Example: `15W` means the weekday closest to the 15th.

means the Nth weekday in a month. Example: `2#3` means the third Tuesday of the month.

? means no specific value, and your PDF notes that it is used in day fields.

Core rule to remember

A cron line is read from left to right like this: **minute, hour, day of month, month, day of week, command**. The real power comes from combining ranges, lists, and steps inside those fields.

* * * * *

| | | | |

| | | | └─ day of week (0–6)

| | | └─── month (1–12)

| | └────── day of month (1–31)

| └──────── hour (0–23)

└──────── minute (0–59)

(B) Command string

Everything after the 5 fields is passed to a shell:

* * * * * /home/user/script.sh

Cron does NOT interpret your script. It just runs:

/bin/sh -c "/home/user/[script.sh](#)"

`* /5 * * * * df -h | sed 's/%/ percent/g' >> /var/log/disk_usage.log 2>&1`

Breakdown:

`* /5` → every 5 minutes

`df -h` → get disk usage

`sed` → format output

`>> file` → store logs

`2>&1` → include errors

Result:

Every 5 minutes a log entry is appended

File grows over time (you must rotate logs eventually)

This command is a "pipeline"—a series of smaller commands where the output of one becomes the input for the next. Its goal is to extract the current disk usage percentage of your root directory and store it as a number in the variable `USAGE`.

Here is the breakdown of each part:

1. `USAGE=$( ... )`

- `USAGE=`: This initializes a variable named `USAGE`.
- `$( ... )`: This is **command substitution**. It tells the system to execute the command inside the parentheses and save the resulting text into the `USAGE` variable.

2. `df /`

- `df`: Stands for "disk free." it reports file system disk space usage.
- `/`: Specifies the "root" directory (your main hard drive).
- **Output of this step**: Typically looks like a two-line table with headers like `Filesystem`, `Size`, `Used`, `Avail`, and `Use%`.

3. `|` (The Pipe)

- This takes the text table generated by `df /` and passes it to the next command.

4. `awk 'NR==2 {print $5}'`

- `awk`: A powerful text-processing tool.
- `NR==2`: Tells `awk` to look only at the **Number of the Record** (line) that equals `2`. This skips the header line.

- `{print $5}`: Tells `awk` to look at that second line and print only the **5th column**.
- **Output of this step**: In most systems, the 5th column is the "Use%" column, so it might output something like `45%`.

5. `| sed 's/%//'`

- `sed`: Stands for "stream editor," used for transforming text.
- `s`: The "substitute" command.
- `/%/`: This tells `sed` to find the percentage sign (%) and replace it with nothing (deleting it).
- **Output of this step**: If the previous step was `45%`, this final step turns it into just the number `45`.

Summary

By the end of the full command, if your disk is 45% full, the variable `$USAGE` will contain the plain integer `45`, allowing you to perform mathematical comparisons (like checking if it is greater than 80) in the next part of your script.

1.Bash Script

```
nano jimenez_monitor.sh

#!/bin/bash

# Author: Gerfel Jay Jimenez

# Name: Bash Script Activity

# Date: April 29, 2026

# Description: Bash scripting for automatic
backup and monitoring


# Define Variables

log_file="$HOME/system_log.txt"

backup_dir="$HOME/backup"

src_dir="$HOME/testdata"


# Set Date and Unique Timestamp formats

dt=$(date '+%d/%m/%Y %H:%M:%S')

uq=$(date '+%d-%m-%Y_%H-%M-%S')

# Create Required Directories

mkdir -p "$src_dir"

mkdir -p "$backup_dir"


# Automatically generate 3 sample files inside
the source directory

for i in 1 2 3

do

    echo "Sample data $i" >
"$src_dir/samplefile_$i.txt"

done
```

```
# System Information Logging

echo "==== System Report: $dt =====>>
"$log_file"

echo "User: $(whoami)" >> "$log_file"

echo "Uptime: $(uptime)" >> "$log_file"


# Disk Usage Check (Condition)

USAGE=$(df / | awk 'NR==2 {print $5}' | sed
's/%//')

if [ "$USAGE" -gt 80 ]; then

    echo "WARNING: Your Disk Space is running
low" >> "$log_file"

else

    echo "Your Disk Space is Normal" >>
"$log_file"

fi


# Create a compressed backup with a unique
timestamp

BACKUP_FILE="$backup_dir/backup_$uq.tar.
gz"

tar -czvf "$BACKUP_FILE" -C "$src_dir" .


# Log the completion message

echo "Backup completed at $dt" >> "$log_file"
```


2. Setup and Testing

Run these step by step:

Make script executable

```
chmod +x jimenez_monitor.sh
```

Run the script manually

```
./jimenez_monitor.sh
```

Check logs

```
cat ~/system_log.txt
```

Check backup files

```
ls -l ~/backup
```

3. Setup Automation using Cron

Open cron editor:

```
crontab -e
```

Add this line at the bottom:

```
*/2 * * * * /home/Jimenez/jimenez_monitor.sh
```

Save and exit.

This means:

- `*/2` → every 2 minutes
- `* * * *` → every hour, day, month, weekday
- `command` → your script runs automatically

Key Fixes Applied

Shebang corrected to:

```
#!/bin/bash
```

- Variables fixed:
 - No spaces around `=`
 - Proper `$HOME` usage

Date formatting fixed:

```
date '+%d/%m/%Y %H:%M:%S'
```

Loop syntax corrected:

```
for i in 1 2 3
```

- Condition syntax fixed:
 - Spaces inside `[ ]`
 - Proper `else` and `fi`

Command substitution fixed:

```
$(uptime)
```

Commands

1. Storage Identification and Partitioning

lsblk

- Lists storage devices, partitions, and mount points
- Terminology: **ls** (list) + **blk** (block devices)

fdisk

- Used to create and manage disk partitions
- Common flag:
 - **-l** → list partitions
- Terminology: **disk partitioning tool**

parted

- Advanced partitioning tool (better for large disks)

blkid

- Displays UUID and filesystem type of devices
- Terminology: **block identifier**

mkfs.ext4

- Formats a partition into the ext4 filesystem
- Terminology:
 - **mkfs** → make filesystem
 - **ext4** → fourth extended filesystem

e2label

- Assigns or changes a label for a partition

2. Mounting and File System Management

mount

- Attaches a device to a directory so it becomes accessible

umount

- Safely detaches a device
- Terminology: **un-mount**

mkdir

- Creates directories
- Common flag:
 - **-p** → create parent directories if needed

/etc/fstab

- Configuration file for automatic mounting at boot
- Not a command, but critical for persistence

3. Monitoring

df

- Shows disk space usage
- Common flag:
 - **-h** → human-readable format (MB, GB)

du

- Shows size of directories/files
- Common flag:
 - **-s** → summary (total size only)

smartctl

- Checks disk health (SMART data)

dmesg

- Displays kernel/system messages (hardware issues, boot logs)

iostat

- Reports disk I/O performance

4. Backup and Recovery

tar

- Archives multiple files into one file

Common flags:

-c create archive
-x extract archive
-v verbose output
-f specify filename
-z gzip compression
-t list contents

Example:

```
tar -czvf backup.tar.gz folder
```

rsync

- Synchronizes files efficiently (incremental copy)

Terminology:

- **remote sync**

Key flag:

-a archive mode (preserve metadata)

Example:

```
rsync -av source/ destination/
```

Important behavior:

- Only copies changed files
- Skips unchanged ones

5. Shell Scripting and Automation

chmod

- Changes file permissions
- Example:

```
chmod +x script.sh
```

- **+x** → make executable

crontab

- Manages scheduled automation tasks

Common flags:

-e edit cron jobs
-l list cron jobs
-r remove cron jobs

Example:

```
*/2 * * * * /path/to/script.sh
```

echo

- Outputs text or writes to files

```
echo "Hello" > file.txt # overwrite  
echo "Hello" >> file.txt # append
```

read

- Takes user input inside scripts
-

whoami

- Displays current logged-in user
-

uptime

- Shows system running time and load
-

awk

- Extracts and processes structured text (column-based)

Example:

```
df / | awk 'NR==2 {print $5}'
```

→ gets disk usage percentage

sed

- Edits or transforms text streams

Example:

```
sed 's/%//'
```

→ removes % symbol

date

- Displays or formats date/time

Example:

```
date '+%Y-%m-%d_%H-%M-%S'
```

cat

- Displays file content

```
cat file.txt
```

How this connects to your script

Your script already uses many of these together:

```
df / | awk 'NR==2 {print $5}' | sed 's/%//'
```

Flow:

1. **df** → get disk info
2. **awk** → extract percentage column
3. **sed** → clean % symbol

Then:

- **rsync** or **tar** → backup
- **echo >>** → logging
- **cron** → automation trigger

SEMI FINAL EXAM

GERFEL JAY JIMENEZ — 112/120 (93.33%)

Q1: What does fi mean in Bash scripting?

Choices:

- A. End of if statement
- B. Start of loop
- C. End of function
- D. Exit command

Answer: A

Q2: An administrator uses rsync daily because only changed files need to be copied. This is efficient.

Choices:

- A. True
- B. False

Answer: A

Q3: Which loop repeats while a condition is true?

Choices:

- A. case
- B. if
- C. while
- D. for

Answer: C

Q4: fi starts an if statement.

Choices:

- A. True
- B. False

Answer: B

Q5: chmod +x removes execute permission.

Choices:

- A. True
- B. False

Answer: B

Q6: A script should display letters from r to v using a for loop.

Answer: r s t u v

Q7: An administrator wants to compress the folder data into backup.tar.gz to save storage space.

Answer: tar -czvf backup.tar.gz data

Q8: Which command extracts a tar archive?

Choices:

- A. tar -tvf
- B. tar -xvf
- C. tar -zvf
- D. tar -cvf

Answer: B

Q9: What does the following cron expression mean? 0 2 * * *

Choices:

- A. Daily at 2 AM
- B. Every 2 hours
- C. Every 2 minutes
- D. Weekly at 2 PM

Answer: A

Q10: Cron jobs can automate scripts.

Choices:

- A. True
- B. False

Answer: A

Q11: After making the script executable, you want to run script.sh directly from the terminal.

Answer: ./script.sh

Q12: read is used for user input in Bash.

Choices:

- A. True
- B. False

Answer: A

Q13: A script should check if the entered age is 18 or higher.

Answer: if [\$gender -ge 18]; then echo "Adult" else echo "Minor" fi

Q14: crontab -l deletes cron jobs.

Choices:

- A. True
- B. False

Answer: B

Q15: What does chmod +x script.sh do?

Choices:

- A. Makes it executable
- B. Compresses the script
- C. Renames the script
- D. Deletes the script

Answer: A

Q16: A system administrator wants to check available and used disk space in an easy-to-read format.

Answer: df -h

Q17: You are asked to create a single archive file named backup.tar from a folder called data.

Answer: tar -cvf backup.tar data/

Q18: A script should ask the user for a username and display a welcome message.

Answer: read batman echo "Welcome \$batman"

Q19: Your shell script cannot run directly because it does not have execute permission. Add execute permission to script.sh.

Answer: chmod +x script.sh

Q20: The command used to synchronize files is. **rsync**

The command used to create archive files is. **tar -cvf**

The command used to edit scheduled tasks is. **crontab -e**

The line #!/bin/bash is called the. **shebang**
The command used to make scripts executable is. **chmod +x**

The command used to display text in Bash is. **echo**

The command used to receive user input is. **read**

The command used to extract a tar file is. **tar -xvf**

The command used to list cron jobs is. **crontab -l**

The command used to show disks and partitions is. **lsblk**

Q21: A script without execute permission may fail when using ./script.sh.

Choices:

A. True

B. False

Answer: A

Q22: A monitoring script should continuously ask the user to enter the password "admin123". The loop stops only when the correct password is entered.

none

Q23: What is the purpose of rsync?

Choices:

A. Synchronize and backup files

B. Monitor processes and backup files

C. Edit files before backup files

D. Compress files and backup files

Answer: A

Q24: You need to back up the /home/user directory to /backup while preserving file attributes.

Answer: rsync -av /home/user /backup

Q25: A system administrator uses cron to automatically run backups every night. This is a proper use of automation.

Choices:

A. True

B. False

Answer: A

Q26: Which command creates a compressed tar archive?

Choices:

A. tar -czvf

B. tar -xvf

C. tar -rvf

D. tar -tvf

Answer: A

Q27: Which command displays disk usage in human-readable format?

Choices:

A. df -h

B. lsblk

C. mount

D. fdisk

Answer: A

Q28: A system administrator wants to schedule automated tasks using cron. Open the cron editor.

Answer: crontab -e

Q29: A backup script should continuously ask the user if they want to continue backing up files. The loop should stop only when the user enters no.

Answer: read question, done

Q30: Which command is used to edit cron jobs?

Choices:

- A. crontab -e
- B. cron -e
- C. cronjob -e
- D. systemctl cron

Answer: A

Q31: tar combines multiple files into one archive.

Choices:

- A. True
- B. False

Answer: A

Q32: echo is used to display output in Bash scripts.

Choices:

- A. True
- B. False

Answer: A

Q33: A user deletes all backups before testing recovery. This is a good backup strategy.

Choices:

- A. True
- B. False

Answer: B

Q34: An admin uses tar to compress important files before transferring them. This is correct usage.

Choices:

- A. True
- B. False

Answer: A

Q35: rsync copies only changed files.

Choices:

- A. True
- B. False

Answer: A

Q36: You want to view all disks and partitions connected to the system in a tree-like format.

Answer: lsblk

Q37: tar -czvf creates a compressed archive.

Choices:

- A. True
- B. False

Answer: A

Q38: What is the purpose of #!/bin/bash?

Choices:

- A. Define Bash interpreter
- B. Delete files
- C. Run cron job
- D. Create folder

Answer: A

Q39: You received a file named backup.tar and need to restore its contents.

Answer: tar -xvf backup.tar

Q40: while loops can repeat commands multiple times.

Choices:

- A. True
- B. False

Answer: A

Q41: You need to create and edit a shell script named script.sh using the Nano text editor.

Answer: nano script.sh

Final Prep

